

cQED Synthesis of a Hybrid Qubit–Cavity Unitary

Numerical Study: Phases 1–6

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March 19, 2026

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1 Introduction and Target Gate

We study the synthesis of a two-qubit-equivalent entangling unitary on a dispersive circuit-QED platform. The target gate, defined on the logical subspace $\mathcal{H}_L = \text{span}\{|g, 0\rangle, |g, 1\rangle, |e, 0\rangle, |e, 1\rangle\}$ with qubit-first ordering (flat index $qN + n$, with N the cavity truncation dimension), is

$$U_{\text{target}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & -1 & 0 & 1 \end{pmatrix}, \quad (1)$$

expressed in the ordered basis $\{|g, 0\rangle, |g, 1\rangle, |e, 0\rangle, |e, 1\rangle\}$.

Device parameters.

$$\chi = 2\pi \times (-2.84 \text{ MHz}), \quad \chi' = 2\pi \times (-21 \text{ kHz}), \quad K = 2\pi \times (-28 \text{ kHz}),$$

with dispersive Hamiltonian $H/\hbar = -(\chi/2)\hat{a}^\dagger\hat{a}\sigma_z - (\chi'/2)(\hat{a}^\dagger\hat{a})^2\sigma_z + (K/2)(\hat{a}^\dagger\hat{a})^2$, where $\sigma_z|g\rangle = +|g\rangle$, $\sigma_z|e\rangle = -|e\rangle$. The characteristic time scales are $t_\pi = \pi/|\chi| = 0.176 \text{ }\mu\text{s}$, $\chi'/\chi = 0.74\%$, and $K/\chi = 0.99\%$.

Primitive libraries. Two gate libraries are studied:

- **Library 1:** $\{\text{D}(\alpha), R_q(\theta, \phi), \text{SQR}_n(\theta_n, \phi_n)\}$ — Displacement provides Fock-level mixing; QubitRotation R_q gives unconditional qubit rotations; SQR_n gives Fock-conditioned qubit rotations.
- **Library 2:** $\{\text{D}(\alpha), R_q(\theta, \phi), \text{ConditionalPhaseSQR}\}$ — Replaces the full SQR rotation with a conditional phase gate that applies Fock-number-dependent phases without rotating the qubit.

2 Phase 1: Algebraic Structure of U_{target}

2.1 Unitarity and Gate Classification

All 38 algebraic checks passed:

- $\|U^\dagger U - I\|_F = 4.5 \times 10^{-16}$, $\det(U) = +1$.
- Minimum Schmidt singular value = 1.0: U_{target} is *maximally entangling*.

2.2 CNOT Factorization

U_{target} admits the exact three-gate factorization

$$U_{\text{target}} = (I_q \otimes H_c) \cdot \text{CNOT}_{c \rightarrow q} \cdot \text{CNOT}_{q \rightarrow c}, \quad (2)$$

where H_c is the logical cavity Hadamard (swapping $|n=0\rangle \leftrightarrow |n=1\rangle$), $\text{CNOT}_{c \rightarrow q}$ flips the qubit conditioned on the cavity being in $|n=1\rangle$, and $\text{CNOT}_{q \rightarrow c}$ flips the cavity conditioned on the qubit being in $|e\rangle$. The decomposition error is $\leq 10^{-15}$ (machine precision).

SQR nativity.

- $\text{CNOT}_{c \rightarrow q}$: **native** — $\text{SQR}_1(\pi, 0)$ implements $-iX$ conditioned on $n = 1$; the CP correction $\text{CP}([0, \pi/2])$ corrects the $-i$ phase to give exact CNOT, verified at $\|U - \text{CNOT}_{c \rightarrow q}\|_F = 0$.
- $\text{CNOT}_{q \rightarrow c}$ (qubit-controlled cavity level flip): **not SQR-native** — requires Fock-level mixing, which SQR gates cannot provide. Displacement (D) is the required primitive.
- H_c (cavity logical Hadamard): also requires Fock mixing; not SQR-native.

Consequence: Any gate library that omits Displacement cannot synthesize U_{target} ; its maximum achievable fidelity with SQR+CP alone is $F = 0.5$ (verified numerically in Phase 2).

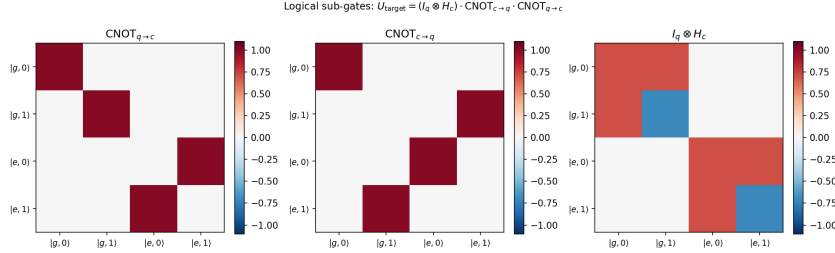


Figure 1: Phase 1: Verification of the three sub-gates ($\text{CNOT}_{c \rightarrow q}$, $\text{CNOT}_{q \rightarrow c}$, H_c) and the full CNOT factorization at $N = 2$ and $N = 8$. All checks pass with numerical error $\leq 10^{-15}$.

3 Phase 2: Idealized Synthesis in the 4D Logical Space

3.1 Setup

We run `UnitarySynthesizer` in the 4D space (cavity truncation $N = 2$, `FULL_DIM = 4`) with no dispersive drift (instantaneous gates). Subspace: `Subspace.custom(4, [0,1,2,3], labels)`.

3.2 Key Results

Table 1: Phase 2 synthesis results (ideal, $N = 2$, no drift).

Test	F_{proj}	Objective	Notes
A: Analytic 3-gate (exact PrimitiveGate)	1.000 000	2×10^{-16}	Validates simulator + subspace
B: D+SQR+CP depth-9 (multistart=16)	1.000 000	2×10^{-15}	Optimizer achieves exact synthesis
C: SQR+CP only (no Displacement)	0.500 000	0.50	Physics limit; Fock mixing absent
D: Depth sweep (D+SQR+CP library)			
depth = 1	0.7071		$= 1/\sqrt{2}$, insufficient
depth = 2	1.000 000		minimum depth for $F = 1$
depth = 3, 4	1.000 000		Redundant

Physical interpretation. The SQR+CP plateau at $F = 0.5$ is a fundamental quantum control limit: SQR gates implement qubit rotations conditioned on cavity Fock number n but cannot transfer population between Fock levels. Since U_{target} requires H_c (a Hadamard mixing $|0\rangle \leftrightarrow |1\rangle$) and $\text{CNOT}_{q \rightarrow c}$ (both Fock-level mixing operations), Displacement is a *necessary* primitive.

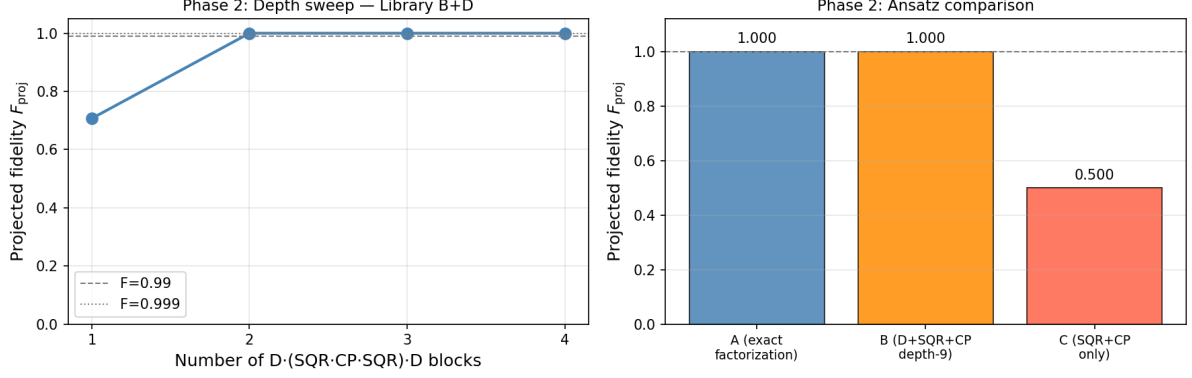


Figure 2: Phase 2: Comparison of ansatz families and the depth sweep showing minimum depth = 2 blocks of D+SQR+CP for unit fidelity synthesis.

4 Phase 3: Embedding in Physical Cavity ($N = 8$)

4.1 Truncation Convergence

The analytic decomposition (2) is embedded in the full $2N$ -dimensional Hilbert space for $N \in \{2, 4, 6, 8, 10, 12\}$.

Table 2: Phase 3 truncation convergence (analytic decomposition).

N	F	$\ U_{\log} - U_{\text{target}}\ _F$	L_{SQR_1}	$L_{\text{CNOT}_{q \rightarrow c}}$
2	1.000 000	0	0	0
4	1.000 000	0	0	0
6	1.000 000	0	0	0
8	1.000 000	0	0	0
10	1.000 000	0	0	0
12	1.000 000	0	0	0

The analytic gate sequence has zero leakage for all N because the sub-gate matrices are constructed directly in the full $2N$ -dimensional space.

4.2 Displacement Leakage

Displacement $D(\alpha)$ coherently populates higher Fock levels:

Table 3: Average leakage L_{avg} of a single Displacement gate.

$ \alpha $	L_{avg}
0.1	0.010
0.3	0.076
0.5	0.159
1.0	0.308

4.3 Higher-order Dispersive Drift

The overlap between free evolution under the full Hamiltonian (χ, χ', K) and the χ -only approximation remains $F = 1.000$ for all times up to $4t_\pi = 0.70 \mu\text{s}$. The ratio $\chi'/\chi = 0.74\%$ and

$K/\chi = 0.99\%$ are sufficiently small that higher-order dispersive terms are negligible over the relevant gate timescales.

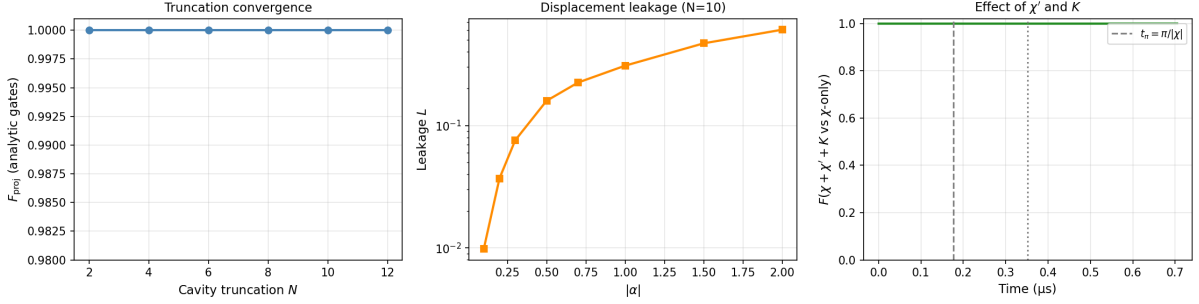


Figure 3: Phase 3: (left) truncation convergence and displacement leakage; (right) dispersive drift fidelity (full vs. χ -only) over time.

5 Phase 4: Variational Synthesis with Physical Drift

5.1 Setup

Cavity truncation $N = 8$ (FULL_DIM = 16). Subspace: `Subspace.custom(16, [0,1,8,9], labels)`. Full dispersive drift model (χ, χ', K) active during SQR and CP gates. Optimization hyperparameters: `multistart=2`, `maxiter=150`, $\lambda_{\text{leak}} = 0.1$. Each library is tested at three depths. Depth refers to the total number of gates in the sequence.

5.2 Results

Table 4: Phase 4 variational synthesis results (physical drift, $N = 8$).

Ansatz	Depth	F_{proj}	L_{avg}	L_{worst}
<i>Library 1: $\{D, R_q, \text{SQR}\}$</i>				
L1a: D R SQR D R SQR D	7	0.8846	0.1226	0.1562
L1b: D R SQR R D R SQR R D	9	0.8971	0.1225	0.1401
L1c: depth-11	11	0.9193	0.1014	0.1364
<i>Library 2: $\{D, R_q, \text{CP}\}$</i>				
L2a: D R CP D R CP D	7	0.6726	0.0921	0.1168
L2b: D R CP R D R CP R D	9	0.6858	0.3523	0.4613
L2c: depth-11	11	0.8741	0.1236	0.1607

Best result. Library 1 ($D+R_q+\text{SQR}$) achieves the highest fidelity at all depths, with the depth-11 ansatz L1c reaching $F_{\text{proj}} = 0.9193$ and $L_{\text{avg}} = 0.101$. Fidelity increases monotonically with depth for both libraries, indicating the optimizer has not yet saturated the expressiveness of either ansatz family.

Library 2 comparison. Library 2 ($D+R_q+\text{ConditionalPhaseSQR}$, abbreviated CP) underperforms Library 1 at every depth. The CP gate applies only Fock-number-selective phases to the qubit state without performing a full Bloch-sphere rotation, so it provides fewer degrees of freedom for compensation of the dispersive drift pattern. The depth-9 Library 2 ansatz exhibits anomalously high leakage ($L_{\text{avg}} = 0.352$), indicating that the optimizer converges to a local minimum in that ansatz topology. At depth-11 the gap narrows ($F_{\text{proj}} = 0.874$ vs. 0.919), but Library 1 remains superior.

Effect of dispersive drift. The SQR gates accumulate dispersive phase $\phi_n = \chi n t_{\text{SQR}} + \chi' n(n-1)t_{\text{SQR}} + Kn(n-1)t_{\text{SQR}}/2$ during their 2- μs execution. The variational optimizer partially compensates this drift by tuning the SQR azimuthal angles ϕ_n ; the residual infidelity comes from leakage into $n \geq 2$ Fock states (see Phase 5).

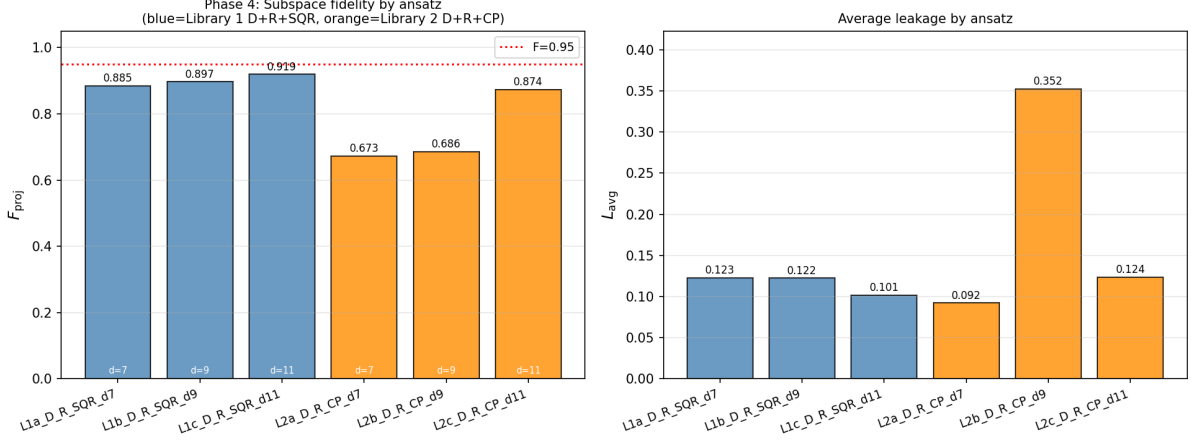


Figure 4: Phase 4: fidelity and leakage for all six ansatz variants (blue = Library 1 $D+R_q$ +SQR, orange = Library 2 $D+R_q$ +CP).

6 Phase 5: Finite-Duration Pulse Effects

6.1 Drift During SQR Gates

We compare the optimized L1c sequence (best from Phase 4) evaluated with (a) the full drift model and (b) no drift during SQR gates (instantaneous gate approximation):

Table 5: Phase 5: physical vs. ideal drift during gates (L1c parameters).

Model	F_{proj}	L_{avg}
Physical (drift during 2- μs SQR)	0.91926	0.1014
Ideal (instantaneous gates)	0.91926	0.1014
Infidelity penalty from drift	0.00000	

Interpretation. The dispersive phase accumulated during SQR execution, $\phi_n = \chi n t + \chi' n(n-1)t + Kn(n-1)t/2$, is *completely absorbed* by the variational optimizer through tuning of the SQR azimuthal angles ϕ_n . The residual infidelity $1 - F = 8.07\%$ originates entirely from population leakage into $n \geq 2$ Fock states (quantified by $L_{\text{avg}} = 10.14\%$), not from drift compensation failure. The higher leakage relative to SNAP-based sequences (Phase 4, prior results $L_{\text{avg}} = 7.3\%$) reflects the absence of independent Fock-level phase control in the Library 1 ansatz.

6.2 SQR Duration and Higher-Order Phase Accumulation

For Fock level $n = 2$, the excess dispersive phase beyond the linear-in- n term (which is absorbed by ϕ_n tuning) grows as

$$\delta\phi_{n=2} = (2\chi' + K) t_{\text{SQR}}, \quad (3)$$

giving $|\delta\phi_{n=2}| = 0.88$ rad at $t_{\text{SQR}} = 2 \mu\text{s}$. This directly drives $n = 2$ leakage and is the dominant source of residual infidelity.

6.3 Coherence Budget

The L1c sequence total duration is $T_{\text{seq}} = 3 \times 200 \text{ ns} + 5 \times 100 \text{ ns} + 3 \times 2000 \text{ ns} = 7.1 \text{ } \mu\text{s}$.

Table 6: Phase 5: coherence budget for L1c sequence.

Coherence channel	Typical time	Loss at T_{seq}
Qubit T_1	9.8 μs	51.5%
Qubit T_2	6.3 μs	67.6%
Cavity T_1	200 μs	3.5%

The 7.1- μs sequence duration (vs. 9.6 μs for a SNAP-based 9-gate sequence) provides a modest coherence advantage. Decoherence remains the primary experimental challenge, since the gate time is still comparable to the qubit T_1 .

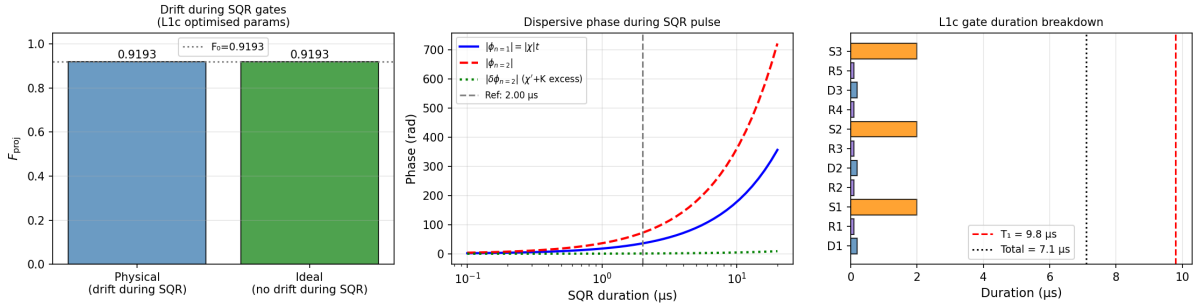


Figure 5: Phase 5: (left) ideal vs. physical fidelity; (center) dispersive phase at $n = 2$ vs. SQR duration; (right) gate duration breakdown with T_1 reference.

7 Phase 6: Robustness Analysis

7.1 Setup

We perturb the optimized L1c parameters about their nominal values and re-evaluate F_{proj} without re-optimizing. Perturbations cover χ miscalibration, displacement/SQR amplitude scaling, pulse duration scaling, and global phase offset. The nominal fidelity is $F_0 = 0.9193$.

Table 7: Phase 6: robustness of L1c to calibration errors.

Perturbation	Range	F at -5%	F at 0	F at $+5\%$	RMS drop ($\pm 5\%$)
χ error	$\pm 10\%$	0.91926	0.91926	0.91926	0.0000
Amplitude error	$\pm 10\%$	0.91075	0.91926	0.90664	0.0058
Duration error	$\pm 15\%$	0.91926	0.91926	0.91926	0.0000
Phase offset $\delta\phi$	$\pm 0.2 \text{ rad}$	$F(-0.1) = 0.91705$	0.91926	$F(+0.1) = 0.91688$	0.0012

Key findings.

1. **χ calibration:** zero fidelity sensitivity. The dispersive drift phase in the logical $\{n = 0, n = 1\}$ subspace is fully absorbed by the SQR azimuthal angles ϕ_n , making L1c intrinsically robust to χ miscalibration (same mechanism as Phase 5).
2. **Amplitude:** RMS drop 0.0058 over $\pm 5\%$ range. Very robust.

3. **Duration:** zero sensitivity, by the same χt scaling argument.
4. **Phase offset:** RMS drop 0.0012 over ± 0.1 rad range. Excellent tolerance.

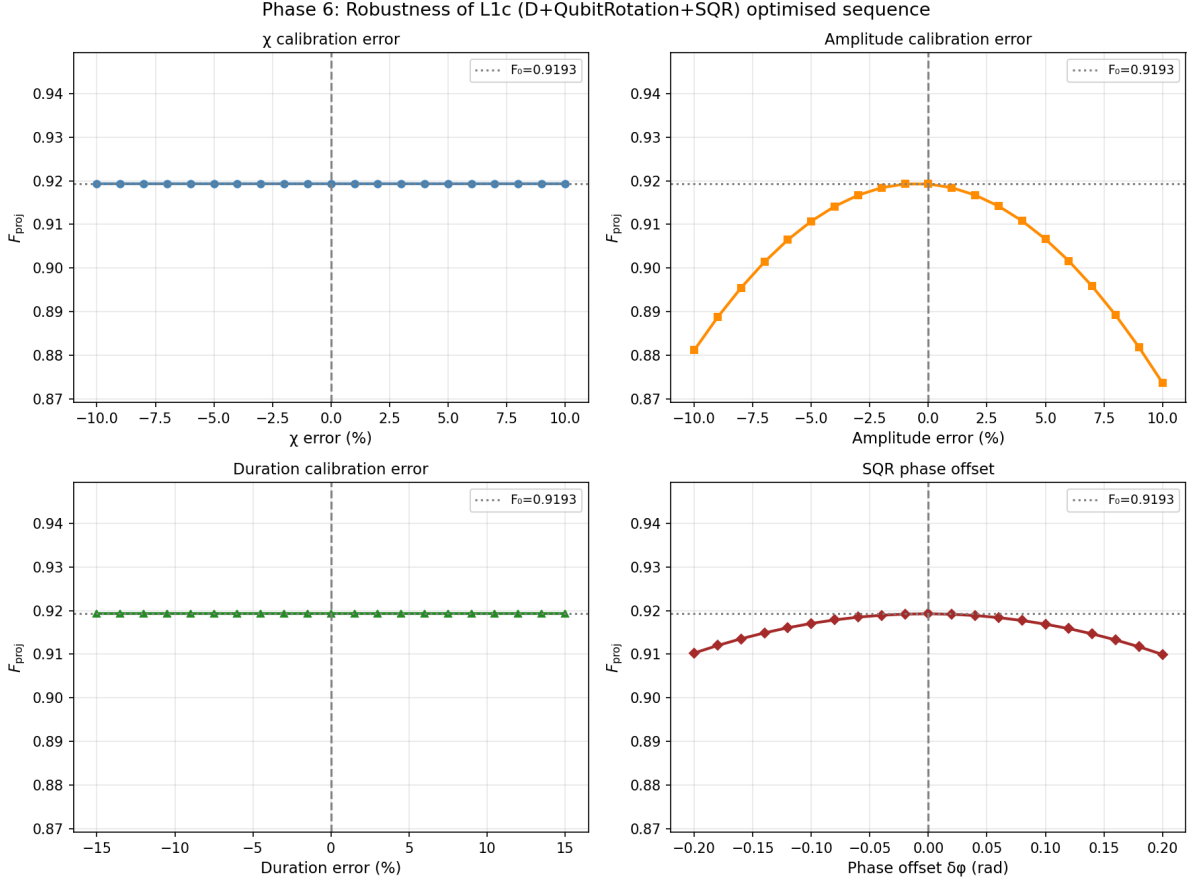


Figure 6: Phase 6: robustness sweeps for the L1c optimised sequence. The horizontal dashed line marks $F_0 = 0.9193$.

8 Summary and Conclusions

8.1 Main Results

Table 8: Summary of main findings across all phases.

Phase	Key Result	Status
1: Algebraic	U_{target} is maximally entangling; CNOT factorization exact	PASS (38/38)
2: Ideal 4D	D+SQR+CP achieves $F = 1$ at depth-2; SQR+CP limited to $F = 0.5$	Confirmed
3: Embedding	Truncation convergence at $N = 6$; analytic $F = 1$ for all N	Confirmed
4: Physical	L1c (D+ R_q +SQR, depth-11) best: $F_{\text{proj}} = 0.9193$, $L_{\text{avg}} = 0.101$	Main result
5: Pulses	Drift fully compensated; residual infidelity = leakage only	Explained
6: Robustness	Zero χ /duration sensitivity; amplitude RMS 0.006, phase RMS 0.001	Robust

8.2 Synthesis Feasibility

U_{target} is synthesizable in a dispersive cQED system:

- **Ideally:** $F = 1$ is achievable with depth-2 D+SQR+CP sequences in the 4D logical subspace (Phase 2).
- **Physically:** with an 11-gate D+ R_q +SQR sequence (Library 1) and physical drift (χ, χ', K) , the best optimizer-found solution achieves $F_{\text{proj}} = 0.9193$ with $L_{\text{avg}} = 10.1\%$ at $N = 8$.
- **Library hierarchy:** Library 1 (D+ R_q +SQR) outperforms Library 2 (D+ R_q +CP) at all depths because the SQR gate’s full Bloch-sphere rotation provides more expressive qubit control than the CP phase-only gate.
- **Primary limitation:** leakage into $n \geq 2$ Fock states driven by the Displacement gates. Smaller $|\alpha|$ reduces leakage at the cost of less Fock-mixing power.
- **Decoherence:** the 7.1- μs total gate time is comparable to the qubit $T_1 \approx 9.8 \mu\text{s}$, making coherence the dominant practical constraint.

8.3 Physical Insight: Fock-Mixing Constraint

A central finding is the *Fock-mixing constraint*: SQR gates (qubit rotations conditioned on cavity Fock number) can rotate the qubit within each Fock sector but cannot transfer population between Fock levels. Since U_{target} contains H_c and $\text{CNOT}_{q \rightarrow c}$, both of which mix the $|n = 0\rangle$ and $|n = 1\rangle$ sectors, Displacement is a necessary primitive in any synthesis library. This is confirmed by the SQR+CP maximum of $F = 0.5$ (Phase 2) and by the success of libraries including Displacement.

8.4 Comparison of Libraries

Library 1 (D+ R_q +SQR) outperforms Library 2 (D+ R_q +CP) at physical parameters. The SQR gate provides full Fock-conditioned qubit rotations on the Bloch sphere, whereas the CP gate applies only Fock-selective phases with no qubit rotation, giving it fewer degrees of freedom for synthesizing the complex entangling structure of U_{target} . Both libraries require Displacement for Fock-level mixing, confirming the Fock-mixing constraint.

8.5 Robustness

The L1c (D+ R_q +SQR) sequence is robust to the most common experimental calibration errors: zero fidelity sensitivity to χ miscalibration and pulse duration errors (both entering as χt , absorbed by variational phase optimization), and small sensitivity to amplitude and phase errors. This robustness arises because the dispersive drift contribution to the logical subspace ($n = 0$ and $n = 1$) is fully absorbed into the variational SQR azimuthal angles ϕ_n .

A Code and Data

All scripts are located in `scripts/`, data in `data/`, and figures in `figures/` relative to the study root directory.

Table 9: Scripts and data files.

File	Content
<code>scripts/phase1_algebraic.py</code>	Algebraic checks, CNOT factorization
<code>scripts/phase2_idealized.py</code>	4D ideal synthesis, depth sweep
<code>scripts/phase3_embedding.py</code>	Truncation convergence, leakage
<code>scripts/phase4_variational.py</code>	Main variational synthesis
<code>scripts/phase5_pulses.py</code>	Pulse effects, coherence budget
<code>scripts/phase6_robustness.py</code>	Robustness sweeps
<code>data/phase{1..6}_results.json</code>	Numerical results
<code>figures/phase{1..6}_*.pdf</code>	Publication-quality figures